

Portainer.io: Build, Manage and Maintain Docker Environments

This article introduces Portainer.io – a lightweight UI that allows the user to easily manage Docker hosts or Swarm clusters. Portainer consists of a single container that can run on any Docker engine. It is easy to install and run, requiring just a few commands.



Portainer.io (or just Portainer from now on) is one of the very lightweight and easy-to-use Docker container-monitoring, building and managing tools. As Docker container adoption and usage is picking up very fast, such a lightweight tool was really needed. The Portainer community edition download is in millions, which indicates this tool's fast growth and usage.

Portainer comprises two elements -- the Portainer server and the agent. Both these elements run as lightweight Docker containers on a standalone Docker engine or within a Swarm cluster. Due to the nature of Docker and its robustness, Portainer can be deployed in multiple ways very easily.

Note: 1. For more details, please refer to *portainer.readthedocs.io*. Also, note that Portainer can be deployed across platforms like Windows and Linux (<https://www.portainer.io/installation/>).

2. For purposes of this article, deployment and installation is done on Ubuntu Linux, and only this option will be discussed.

3. The agent is not needed on standalone hosts; however, it does provide additional functionality if used.

Portainer (latest version is 1.20.2) provides users with a detailed overview of their Docker environments, and allows the management of containers, images, networks and volumes. It is simple to deploy -- you are just one/two Docker commands away from running Portainer anywhere. This tool has many advantages over other tools, some of which are listed below:

- 1 It's lightweight (only two to three commands need to be run to install this tool, and the installation image is only around 26-30MB in size).
- 2 It is robust and easy to use.
- 3 It can be used for Docker monitoring and build.
- 4 This tool provides a detailed overview of Docker environments.
- 5 It allows the management of containers, images, networks and volumes.
- 6 It is simple to deploy (can be run from anywhere).
- 7 Complete Docker-container environment can be monitored easily.

Portainer also has:

- Community support
- Enterprise support
- Availability of professional services
- Availability of partner OEM services